Model MEML 75:

Quark and lepton mass matrices from $T^2/\mathbb{Z}_3$ geometry with axion-like (ALP) dark matter

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Abstract

We present model MEML 75, which parametrizes the fermion sector of the Standard Model through the geometry of the six-dimensional orbifold $T^2/\mathbb{Z}_3$ with three independent tori. The model fits **13 fermion observables** (9 quarks + 4 leptons) with **16 free parameters** ($N_{\rm dof} = 13 - 16 = -3$).

The main testable prediction: axion-like dark matter with $f_a = 1.9801 \times 10^{11}$ GeV, $m_a = 28.407$ μeV, $\nu_a = 6.8688$ GHz, and ALP-photon coupling $g_{a\gamma\gamma} = +4.38 \times 10^{-15}$ GeV^{-1} (DFSZ) or -1.13×10^{-14} GeV^{-1} (KSVZ), testable at ADMX/HAYSTAC (0.5–10 GHz).

Quark sector. A global fit with loss function $\chi^2 = \sum_i ((O_i - O_i^{\rm target})/\sigma_i)^2$ yields $\chi^2 = 6.195 \times 10^{-20}$ at $\tau_u = -0.1577944085 + 2.3851782590\,i$, $\tau_d = +0.3693905617 + 1.5410496057\,i$.

Lepton sector. Using a third torus τ_l, a combination of A_4 modular forms $Y^{(2)}$ and $Y^{(4)}$ yields $\chi^2 < 10^{-16}$ for the PMNS observables at $\tau_l = 0.385257 + 0.958275\,i$. The model predicts $\theta_{12} = 33.41°$, $\theta_{13} = 8.54°$, $\theta_{23} = 49.10°$, and $J_{cp} = -0.018$.

Verdict: the model is *phenomenological*. It interpolates fermion observations with high precision.

1 Introduction

Modular models of fermion masses [4, 5] build the mass hierarchy and mixing from compactification geometry. In model MEML 75, quarks and leptons are localized on three independent tori $T^2/\mathbb{Z}_3$ with moduli τ_u, τ_d, and τ_l. The mass matrices are built from A_4 modular forms of weights $(2, 4, 6)$.

2 Geometry of $T^2/\mathbb{Z}_3$

The orbifold $T^2/\mathbb{Z}_3$ is defined by:

$$z = x + \tau y, \quad ds^2 = dx^2 + dx\,dy + dy^2, \tag{2.1}$$

where $\tau = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$. The Z_3-action $z \to \omega z$ (with $\omega = e^{2\pi i/3}$) has three fixed points, providing a topological explanation for three generations.

3 Bulk coupling α_G (postulate)

$$\alpha_G = \frac{2}{3\pi} = 0.2122065908. \tag{3.1}$$

Geometric motivation: $2/3$ is the volume fraction of the fundamental domain of the $\mathbb{Z}_3$-orbifold; $1/\pi$ is the normalization of the theta-function period.

4 Suppression factor ε (ansatz)

$$\varepsilon = \left(\frac{\alpha_G}{4\pi}\right)^2 = \frac{1}{36\pi^4} = 2.851662 \times 10^{-4}. \tag{4.1}$$

5 Conditionally derived quantities

$$m_{KK} = \frac{M_{Pl}^{\text{red}}}{\sqrt{\mathcal{I}}\, e^{\pi^2}} = 2.0838 \times 10^{14}\ \text{GeV}, \tag{5.1}$$

$$M_6 = \left(3(M_{Pl}^{\text{red}})^2 m_{KK}^2\right)^{1/4} = 2.9646 \times 10^{16}\ \text{GeV}, \tag{5.2}$$

at $\mathcal{I} = 0.3653$.

6 Axion-like (ALP) sector

6.1 ALP parameters

$$f_a = M_{Pl}^{\text{red}}\, \varepsilon^2 = 1.9801 \times 10^{11}\ \text{GeV}, \tag{6.1}$$

$$m_a = \frac{(\chi^{1/4})^2}{f_a} = 28.407\ \mu\text{eV}, \tag{6.2}$$

$$\nu_a = \frac{m_a c^2}{h} = 6.8688\ \text{GHz}. \tag{6.3}$$

Here $\chi^{1/4} \approx 75.0$ MeV is the fourth root of the topological susceptibility of QCD. We emphasize that no explicit $U(1)_{PQ}$ symmetry is introduced; the object is an ALP.

6.2 ALP-photon coupling

$$g_{a\gamma\gamma} = \frac{\alpha_{\rm EM}}{2\pi f_a}\left(\frac{E}{N} - 1.92\right). \tag{6.4}$$

At $f_a = 1.9801 \times 10^{11}$ GeV: $\alpha_{\rm EM}/(2\pi f_a) = 5.866 \times 10^{-15}$ GeV^{-1}.

Table 1: Prediction for $g_{a\gamma\gamma}$.

Scenario	E/N	$g_{a\gamma\gamma}$, GeV^{-1}
KSVZ	0	-1.126×10^{-14}
DFSZ	8/3	$+4.379 \times 10^{-15}$

6.3 ALP dark matter

$$\Omega_a h^2 \approx 0.12\,\theta_i^2\, f_{\rm anh}(\theta_i)\left(\frac{f_a}{3\times 10^{11}}\right)^{7/6}, \tag{6.5}$$

where $f_{\rm anh}(\theta_i) = [\ln(e/(1-(\theta_i/\pi)^2))]^{7/6}$ is the anharmonicity factor. Numerical T-effects introduce a +12.4% correction, leading to agreement with Planck at $\theta_i = 1.1162$.

7 Quark sector (11 parameters)

7.1 Matrix structure

$$M_u = \begin{pmatrix} a_u Y_1^{(2)} & b_u Y_3^{(4)} & c_u Y_2^{(6)} \\ a_u Y_3^{(2)} & b_u Y_2^{(4)} & c_u Y_1^{(6)} \\ a_u Y_2^{(2)} & b_u Y_1^{(4)} & c_u Y_3^{(6)} \end{pmatrix},\ M_d = \begin{pmatrix} a_1 Y_1^{(6)} & b Y_3^{(2)} & c Y_2^{(2)} \\ a_1 Y_3^{(6)} & b Y_2^{(2)} & c Y_1^{(2)} \\ a_2 Y_2^{(2)} & b Y_1^{(2)} & c Y_3^{(4)} \end{pmatrix}. \tag{7.1}$$

7.2 Target values

Table 2: Target values (PDG 2024).

Observable	$O_i^{\rm target} \pm \sigma_i$
m_u/m_t	$(1.27 \pm 0.30) \times 10^{-3}/172.0$
m_c/m_t	$(0.620 \pm 0.020)/172.0$
m_d/m_b	$(2.70 \pm 0.20) \times 10^{-3}/2.860$
m_s/m_b	$(55.0 \pm 5.0) \times 10^{-3}/2.860$
m_b/m_t	$(2.860 \pm 0.030)/172.0$
$\lvert V_{us}\rvert$	0.22501 ± 0.00068
$\lvert V_{cb}\rvert$	0.04053 ± 0.00042
$\lvert V_{ub}\rvert$	0.00382 ± 0.00020
$\lvert J\rvert$	$(3.080 \pm 0.130) \times 10^{-5}$

7.3 Results

Independent optimization yields $\chi^2 = 6.195 \times 10^{-20}$ at the moduli given in the abstract. The coefficients are:

$$c^{\rm fit} \approx (5.31 \times 10^{-5},\ 7.19,\ 0.039,\ 2.43 \times 10^{-4},\ 0.503,\ 1.37 \times 10^{-3},\ 2.22 \times 10^{-3}). \tag{7.2}$$

8 Lepton sector (5 parameters)

The charged lepton masses are taken diagonal, while the neutrino mass matrix M_ν is constructed from the combination of weight-2 and weight-4 modular forms.

8.1 Matrix structure

$$M_\nu = g_1 M_{3_S}(Y^{(2)}) + g_2 M_{3_S}(Y^{(4)}) + d \cdot \mathbb{I}, \tag{8.1}$$

where the triplet contraction $M_{3_S}(Y)$ is given by:

$$M_{3_S}(Y) = \begin{pmatrix} 2Y_1 & -Y_3 & -Y_2 \\ -Y_3 & 2Y_2 & -Y_1 \\ -Y_2 & -Y_1 & 2Y_3 \end{pmatrix}. \tag{8.2}$$

The diagonal term $d \cdot \mathbb{I}$ breaks the $\mu - \tau$ symmetry.

8.2 Results

Independent optimization yields:

$$\boxed{\chi^2 < 10^{-16} \quad \text{at} \quad \tau_l = 0.385257 + 0.958275\,i,\ g_1 = -19.9981,\ g_2 = -19.4310,\ d = 14.3320.} \tag{8.3}$$

Table 3: PMNS observables (NuFIT 5.2).

Observable	Predicted	Target	Deviation
θ_{12} (solar)	33.41°	$33.41^\circ \pm 0.5^\circ$	$\sim 10^{-9}$
θ_{13} (reactor)	8.54°	$8.54^\circ \pm 0.1^\circ$	$\sim 10^{-9}$
θ_{23} (atmos.)	49.10°	$49.1^\circ \pm 0.5^\circ$	$\sim 10^{-9}$
J_{cp} (CP phase)	-0.0180	-0.018 ± 0.005	$\sim 10^{-9}$

9 Full balance

Table 4: Free parameters (16).

Category	Parameters	Count
Quark moduli	τ_u, τ_d (complex)	4
Quark coefficients	$a_u, b_u, c_u, a_1, a_2, b, c$	7
Lepton modulus	τ_l (complex)	2
Lepton coefficients	g_1, g_2, d	3
Total		**16**

$$N_{\text{dof}} = 13 - 16 = -3. \tag{9.1}$$

9.1 Information criteria

Table 5: AIC/BIC for the fermion sector ($n = 13$).

Model	k	χ^2	AIC	BIC
MEML 75	16	$\approx 10^{-20}$	32.0	41.0
SM	19	0.9	38.9	49.6

10 Open problems

1. **Coefficient degeneracy.** The coefficients c_i are non-unique: χ^2 is invariant under $c_i \to kc_i$ (Appendix D).
2. **Status of α_G.** Postulate. Systematic search yields no exact formula.
3. **Status of ε.** Ansatz; choice of 4π is not derived.
4. **Flatness of $\mathrm{Re}\,\tau$.** $\mathrm{Re}\,\tau$ is soft-flat (Appendix E).
5. **LOO instability.** At $N_{\mathrm{dof}} = -3$, spread of $|V_{ub}|$ from -97% to $+25407\%$.

11 Roadmap

Table 6: Phases of development.

Phase	Task	Priority
I	Mechanism for c_i: holomorphic 6D→4D couplings	Critical
I	Flux stabilization of $\mathrm{Im}\,\tau$ ($\pm 3\%$)	High
II	Reduce parameters in quarks	Medium
III	ADMX, DESI, DUNE, KamLAND-Zen	External

12 Conclusions

What has been done:

1. A joint quark and lepton fit with 16 parameters for 13 observables ($N_{\mathrm{dof}} = -3$), plus a testable axion prediction.
2. Quark sector: $\chi^2 = 6.195 \times 10^{-20}$.
3. Lepton sector: $\chi^2 < 10^{-16}$ with $\tau_l = 0.385257 + 0.958275\,i$. The $\mu - \tau$ symmetry breaking is successfully achieved via the $d \cdot \mathbb{I}$ term.

Verdict. The model is phenomenological. It reproduces observations but does not predict. The coefficient degeneracy shows that the model interpolates 13 observations with 16 parameters.

Data Availability

The datasets and the Python verification script generated during and/or analysed during the current study are available in the Zenodo repository at https://doi.org/10.5281/zenodo.23503366 and in the GitHub repository at https://github.com/kolbasa1972/MEML-75-Complete.

A A_4 modular forms: full q-expansions

Definition: $q = e^{2\pi i\tau}$.

$$Y_1^{(2)} = 1 + \sum_{n=1}^{17} C_1^{(n)} q^n, \tag{A.1}$$

$$Y_2^{(2)} = -6\, q^{1/3} \left(1 + \sum_{n=1}^{11} C_2^{(n)} q^n\right), \tag{A.2}$$

$$Y_3^{(2)} = -18\, q^{2/3} \left(1 + \sum_{n=1}^{10} C_3^{(n)} q^n\right), \tag{A.3}$$

$C_1 = [12, 36, 12, 84, 72, 36, 96, 180, 12, 216, 144, 36, 156, 96, 84, 240, 36]$; $C_2 = [1, 7, 8, 18, 14, 30, 20, 42, 30, 56, 42]$; $C_3 = [1, 2, 5, 6, 12, 8, 14, 10, 22, 16]$.

Convolutions:

$$Y_1^{(4)} = (Y_1^{(2)})^2 - Y_2^{(2)} Y_3^{(2)}, \tag{A.4}$$

$$Y_2^{(4)} = (Y_3^{(2)})^2 - Y_1^{(2)} Y_2^{(2)}, \tag{A.5}$$

$$Y_3^{(4)} = (Y_2^{(2)})^2 - Y_1^{(2)} Y_3^{(2)}, \tag{A.6}$$

$$Y_1^{(6)} = \tfrac{1}{3}\left[2Y_1^{(2)} Y_1^{(4)} - Y_2^{(2)} Y_3^{(4)} - Y_3^{(2)} Y_2^{(4)}\right], \tag{A.7}$$

$$Y_2^{(6)} = \tfrac{1}{3}\left[2Y_3^{(2)} Y_3^{(4)} - Y_1^{(2)} Y_2^{(4)} - Y_2^{(2)} Y_1^{(4)}\right], \tag{A.8}$$

$$Y_3^{(6)} = \tfrac{1}{3}\left[2Y_2^{(2)} Y_2^{(4)} - Y_3^{(2)} Y_1^{(4)} - Y_1^{(2)} Y_3^{(4)}\right]. \tag{A.9}$$

B Numerical fit procedure

The global fit was performed using `differential_evolution` (SciPy): 3 seeds (42, 43, 44); maxiter= 150; popsize= 15; tol= 10^{-8}. Bounds: $\mathrm{Re}\,\tau \in [-0.5, 0.5]$, $\mathrm{Im}\,\tau \in [0.87, 2.5]$, $c_i \in [10^{-8}, 10^2]$.

C Reproduction of τ and χ^2

Table 7: Three independent DE runs (quark sector).

Source	$\mathrm{Re}\,\tau_u$	$\mathrm{Im}\,\tau_u$	$\mathrm{Re}\,\tau_d$	$\mathrm{Im}\,\tau_d$
Run 1	−0.15779	2.38518	+0.36939	1.54105
Run 2	−0.15779	2.38518	+0.36939	1.54105
Run 3	−0.12893	2.38411	−0.37212	1.54102

D Proof of coefficient degeneracy

At fixed τ, multiply all c_i by k:

Table 8: Invariance of χ^2 under $c_i \to k c_i$.

k	χ^2
0.5	6.20×10^{-20}
1.0	6.20×10^{-20}
1.87	1.08×10^{-19}
10.0	3.00×10^{-19}

E Flatness of $\mathrm{Re}\,\tau$

Scan in $\mathrm{Re}\,\tau_u$: width of $\chi^2 < 10$: $[-0.40, +0.00]$ (0.40). Scan in $\mathrm{Re}\,\tau_d$: width of $\chi^2 < 10$: $[+0.30, +0.50]$ (0.20).

F LOO test

For each $j \in \{1, \ldots, 9\}$: remove the j-th observable, predict the j-th. $N_{\mathrm{dof}} = +1$. Results: mean $|V_{ub}|$ +16728%, std 11819%.

G Systematic search for α_G

Tested $N = 392$ combinations. **2 coincidences below** 0.1%**:** $|Y_1^{(6)}|/|Y_3^{(4)}| = 0.212125$ (-0.0385%); $|Y_2^{(6)}|/\pi = 0.212411$ ($+0.0965\%$). Poisson: $\langle n \rangle = 0.78$, $n_{\mathrm{obs}} = 2$; $P(n \geq 2) = 0.185$.

References

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